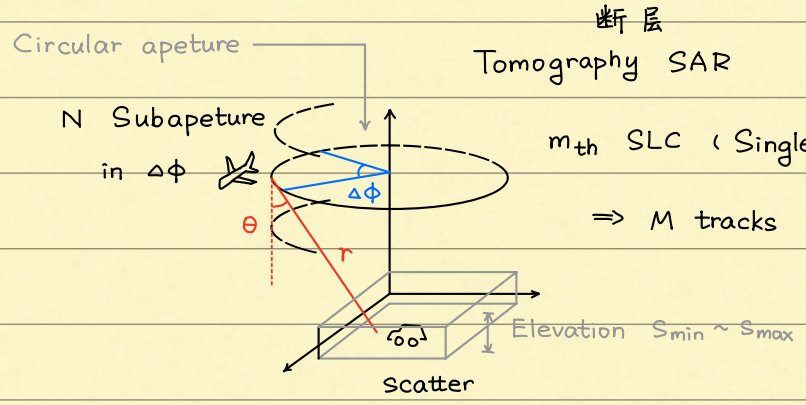
SAR Sythetic Apeture Radar 合成孔径雷达

A Phase Calibration Method Based on Phase Gradient Autofocus for Airborne Holographic SAR Imaging

全息

断层

Tomography SAR



m_{th} SLC (Single Look Complex) 2D-SAR

$\Rightarrow$ M tracks 3D-SAR

For m_{th} track, Given location θ, r

Measurment

$$g(\theta, r, b_m) = \int_{\Delta S} \gamma(s) \cdot e^{-j2\pi \cdot s \cdot \xi_m}$$

Baseline

Reflection Func

Spatial Freq

Elevation 海拔

$$\xi_m = 2b_m / \lambda r$$

$$= \int_{\Delta S} \gamma(s) e^{-j4\pi b_m s / \lambda r}$$

b_m Baseline

基线：阵列间距

Discrete $g(b_m) = \sum_{g=1}^{G} e^{-j4\pi b_m g / \lambda r G} \cdot \gamma_g$ G grid number

$$g_{M\times 1} = \Phi_{M\times G} \gamma_{G\times 1} + e$$

Partial Fourier

γ 可以稀疏

Calibration $g = \alpha_{M\times 1} \odot \Phi \gamma + e$

$$\alpha = [e^{j\vec{\varphi}}]_{M\times 1}$$